



8/13/2026

ASSESSMENT 1

CompTIA A+

A+ - Troubleshooting & Helpdesk

Hardware Components & System
Assembly

✂ Section A – Concept Application

Scenario 1.

- ❖ You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.
- ❖ Question: Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.
 - For a new office workstation, I would first check the system requirements and then select compatible parts from the supplier's list.
 - **Motherboard:** Select a motherboard with the correct **CPU socket**, supported **RAM type and speed**, and suitable **SATA or M.2 slots** for storage.
 - **CPU:** Choose a CPU that matches the motherboard's **socket** and is supported by its chipset.
 - **RAM:** Select RAM with the correct **DDR generation, capacity, and supported speed**.
 - **Storage:** Choose an **SSD or other storage device** that matches the motherboard's available **SATA or M.2 interface**.

To confirm compatibility, I would check the motherboard's specifications and compatibility lists for the CPU and RAM. I would also verify the storage interface and available slots. After assembly, I would check the **BIOS/UEFI and POST** to confirm that the CPU, RAM, and storage are detected correctly.

This ensures that all four components work together properly and the workstation is ready for office use.

Scenario 2.

❖ You are advising a client whose employees mainly run office and productivity applications on their workstations.

❖ Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

➤ For a new office workstation, I would first check the system requirements and then choose compatible components from the supplier's parts list.

➤ **Motherboard:** Select a motherboard with the correct **CPU socket**, supported **RAM type and speed**, and suitable **SATA or M.2 slots** for storage.

➤ **CPU:** Choose a CPU that matches the motherboard's **socket** and is supported by the motherboard.

➤ **RAM:** Select RAM with the correct **DDR type, capacity, and supported speed**.

➤ **Storage:** Choose a storage device that matches the motherboard's available **SATA or M.2 interface**.

To confirm that they will work together, I would check the motherboard specifications and compatibility lists for the CPU and RAM. I would also verify the storage interface and available slots. After assembly, I would check the BIOS/UEFI and POST to make sure the CPU, RAM, and storage are detected correctly.

This confirms that the selected components are compatible and the workstation can work properly.

Scenario 3.

- ❖ **You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.**
- ❖ **Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.**

➤ **Explaining the GPU:**

The **GPU (Graphics Processing Unit)** handles graphics and visual processing for design software. It reduces the workload on the CPU and helps the workstation run smoothly during graphics-intensive tasks.

🛠️ **Explaining the Cooling System:**

The cooling system removes the heat produced by the CPU and GPU during long hours of use. It prevents overheating and helps maintain stable performance and system reliability.

Together, a suitable GPU and effective cooling system help the workstation maintain stable performance during sustained workloads.

Scenario 4.

- ❖ You are assembling a PC from components sourced from three different vendors.
- ❖ Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

- Hardware compatibility is important because all components must fit, communicate, and work together correctly for the computer to function properly.

Why compatibility matters:

Compatible components ensure proper communication between the **motherboard, CPU, RAM, storage, and other hardware**. This helps the system work reliably and prevents installation or boot problems.

Possible issues with incompatible components:

- ⚡ The CPU may not fit or be supported by the motherboard.
- ⚡ The RAM may have an unsupported **DDR type or speed**.
- ⚡ The storage device may not match the available **SATA or M.2 interface**.
- ⚡ The system may **fail to boot**, fail to detect a component, or become unstable.

Therefore, checking compatibility before assembly helps avoid hardware problems and ensures a reliable computer system.

Scenario 5.

- ❖ You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

 - ❖ Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.
- To choose the correct PSU (Power Supply Unit), I would first estimate the total power required by the motherboard, CPU, GPU, storage, and other components. I would then select a PSU with a suitable wattage rating and some extra capacity for reliable operation.

Safety and connection checks:

- ⚙ Switch off and unplug the PSU before making connections.
- ⚙ Check that the PSU is properly mounted and all cables are in good condition.
- ⚙ Connect the required power cables securely to the **motherboard, GPU, and storage devices**.
- ⚙ Make sure every connector is fully seated and connected to the correct socket.
- ⚙ Check that no loose cables or incorrect connections are present before the first boot.

These checks help prevent **power-related damage** and ensure the system can start safely.

Scenario 6.

❖ You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

❖ Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

➤ To confirm that the workstation passes **POST (Power-On Self-Test)**, I would check the following:

➤ Power on the system and confirm that the system starts normally.

➤ Check that there are no **error messages, warning sounds, or diagnostic indicators** during startup.

➤ Enter the **BIOS/UEFI** and verify that the CPU, RAM, storage device, and other required hardware are detected correctly.

➤ Confirm that the system continues to the operating system without any startup error.

If the system does not pass POST, I would first switch off the system and check the power connections, RAM, CPU, and other hardware connections. I would reseat any incorrectly installed component and check for diagnostic or BIOS error messages. After correcting the issue, I would power on the system again and repeat the POST check.

❌ Section B – Practical Coding Tasks

Task 1.

❖ Component Identification & Functions.

DESKTOP PC CORE COMPONENTS – IDENTIFICATION, FUNCTIONS & SPECIFICATIONS

1. MOTHERBOARD

Function: The motherboard connects all components of the system and allows them to communicate. It provides power distribution, data transfer and houses key sockets and slots.

Specifications (where visible): Make/Model: MSI B550M PRO-VDH WIFI
Chipset: AMD B550
Form Factor: Micro ATX
RAM Slots: 4 x DDR4
Expansion Slots: 1 x PCIe 4.0 x16, 1 x PCIe 3.0 x1

2. CPU (PROCESSOR)

Function: The CPU is the brain of the computer. It executes instructions, processes data and controls all operations.

Specifications (where visible): AMD Ryzen 5 5600X
Cores/Threads: 6 Cores / 12 Threads
Base Clock: 3.7 GHz
Max Boost Clock: 4.6 GHz
Socket Type: AM4

3. RAM (MEMORY)

Function: RAM is temporary memory that stores data and instructions currently in use for quick access by the CPU.

Specifications (where visible): Make/Model: Vengeance LPX
Capacity: 16 GB (8GB x 2)
Type: DDR4
Speed: 3200 MHz

4. GPU (GRAPHICS CARD)

Function: The GPU renders images, videos and 3D graphics. It handles visual output and improves performance in games, design and AI tasks.

Specifications (where visible): Make/Model: Gigabyte GeForce RTX 3060
Memory: 12 GB GDDR6
Interface: PCIe 4.0 x16
Outputs: DisplayPort, HDMI

5. STORAGE DEVICE

Function: The storage device stores the operating system, applications and user data permanently.

Specifications (where visible): WD Black SN770
Type: M.2 NVMe SSD
Capacity: 1 TB
Interface: PCIe Gen4 x4

6. COOLING SYSTEM

Function: The cooling system keeps the CPU and other components at safe temperatures by dissipating heat.

Specifications (where visible): Make/Model: Cooler Master Hyper 212 RGB
Type: Air Cooler
Fan Size: 120mm
Lighting: RGB

Note: Specifications are noted as visible on the components provided.

Figure 1 : Desktop PC Core Components - Identification, Functions & Specifications.

❧ Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system.

➤ As shown in Figure 1, the main hardware components and their functions are:

<u>Component</u>	<u>Function</u>
Motherboard	Connects all major components and allows them to communicate with each other.
CPU (Processor)	Executes instructions, processes data, and controls the main operations of the computer.
RAM (Memory)	Temporarily stores data and instructions that are currently being used for quick access.
GPU (Graphics Card)	Processes and renders graphics, videos, and other visual tasks.
Storage Device	Permanently stores the operating system, applications, and user data.
Cooling System	Removes heat from the components and helps maintain a safe operating temperature.

🌀 **Note the make/model or key specification of each component where visible.**

➤ The **make/model and visible specifications** shown in Figure 1 are:

<u>Component</u>	<u>Make/Model or Key Specification</u>
Motherboard	MSI B550M PRO-VDH WIFI, AMD B550 chipset, 4 × DDR4 RAM slots

CPU	AMD Ryzen 5 5600X, 6 Cores / 12 Threads, AM4 socket
RAM	Corsair Vengeance LPX, 16 GB (8 GB × 2), DDR4, 3200 MHz
GPU	Gigabyte GeForce RTX 3060, 12 GB GDDR6, PCIe 4.0 ×16
Storage Device	WD Black SN770, 1 TB, M.2 NVMe SSD, PCIe Gen4 ×4
Cooling System	Cooler Master Hyper 212 RGB, Air Cooler, 120 mm fan

3 Present the findings as a labeled component list or table.

- The components are clearly labelled in Figure 1, and their functions and visible specifications are given above. This makes the hardware easy to identify and understand

Task 2.

❖ Motherboard Ports & Socket Identification.

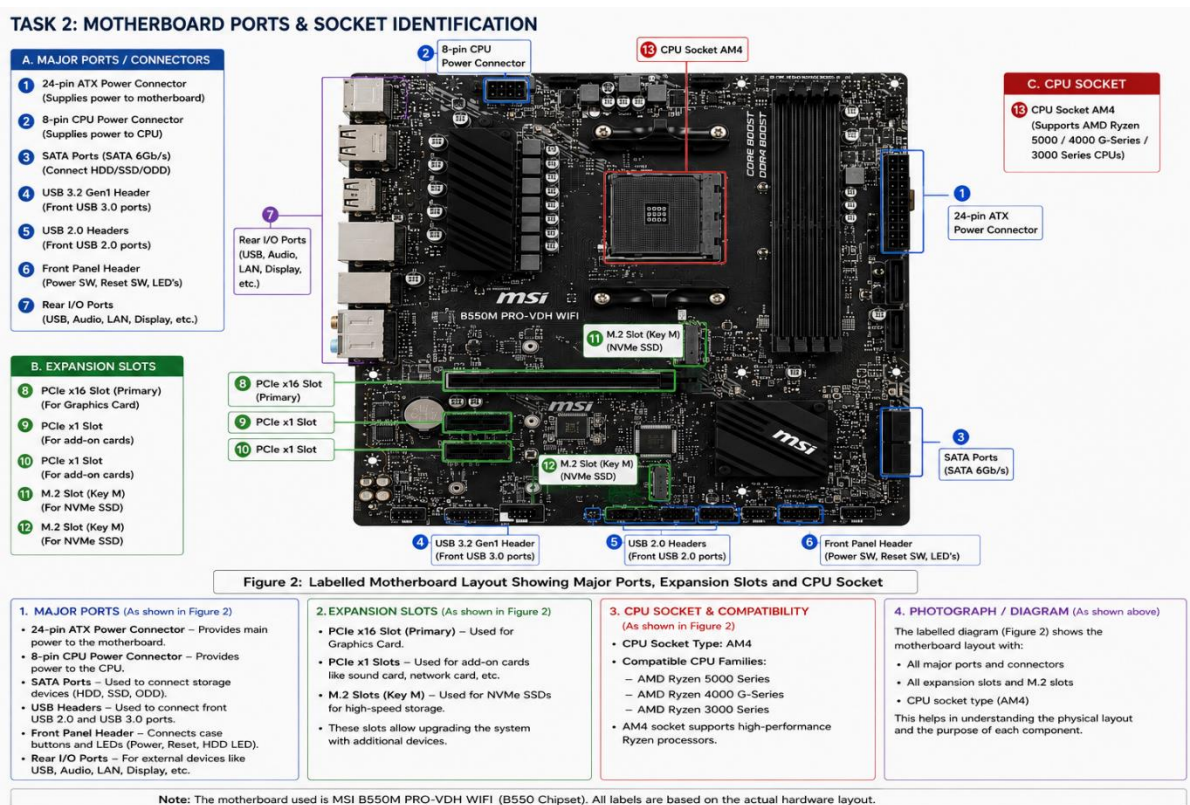


Figure 2: Labelled Motherboard Layout Showing Major Ports, Expansion Slots and CPU Socket.

🌀 Identify and label all major motherboard ports (SATA, power, USB headers, etc.).

➤ As shown in **Figure 2**, the major ports and connectors are:

- **24-pin ATX Power Connector:** Supplies the main power to the motherboard.
- **8-pin CPU Power Connector:** Provides power to the CPU.
- **SATA Ports:** Used to connect SATA storage devices such as HDDs and SSDs.

- **USB Headers:** Connect the front USB ports of the computer case to the motherboard.
- **Front Panel Header:** Connects the power switch, reset switch, and indicator LEDs.
- **Rear I/O Ports:** Provide connections for external devices such as USB, audio, LAN, and display devices.

🔗 Identify all expansion slots (PCIe, M.2, etc.) and their purpose.

- As labelled in Figure 2:

- **PCIe x16 Slot:** Used mainly for installing a graphics card.
- **PCIe x1 Slots:** Used for additional expansion cards such as network or sound cards.
- **M.2 Slots:** Used for installing M.2 NVMe SSDs for high-speed storage.

🔗 Identify the CPU socket type and note which CPU families are compatible with it.

- The motherboard shown in **Figure 2** uses an **AM4 CPU socket**. It supports compatible AMD Ryzen processor families, including **Ryzen 5000 Series, Ryzen 4000 G-Series, and Ryzen 3000 Series**.

🔗 Photograph or diagram the motherboard with labels.

- Figure 2 provides a labelled view of the motherboard, showing its major ports and connectors, expansion slots, M.2 slots, and AM4 CPU socket. This makes the physical layout and purpose of the main components easy to identify.

Task 3.

❖ RAM & Storage Installation.

TASK 3: RAM & STORAGE INSTALLATION

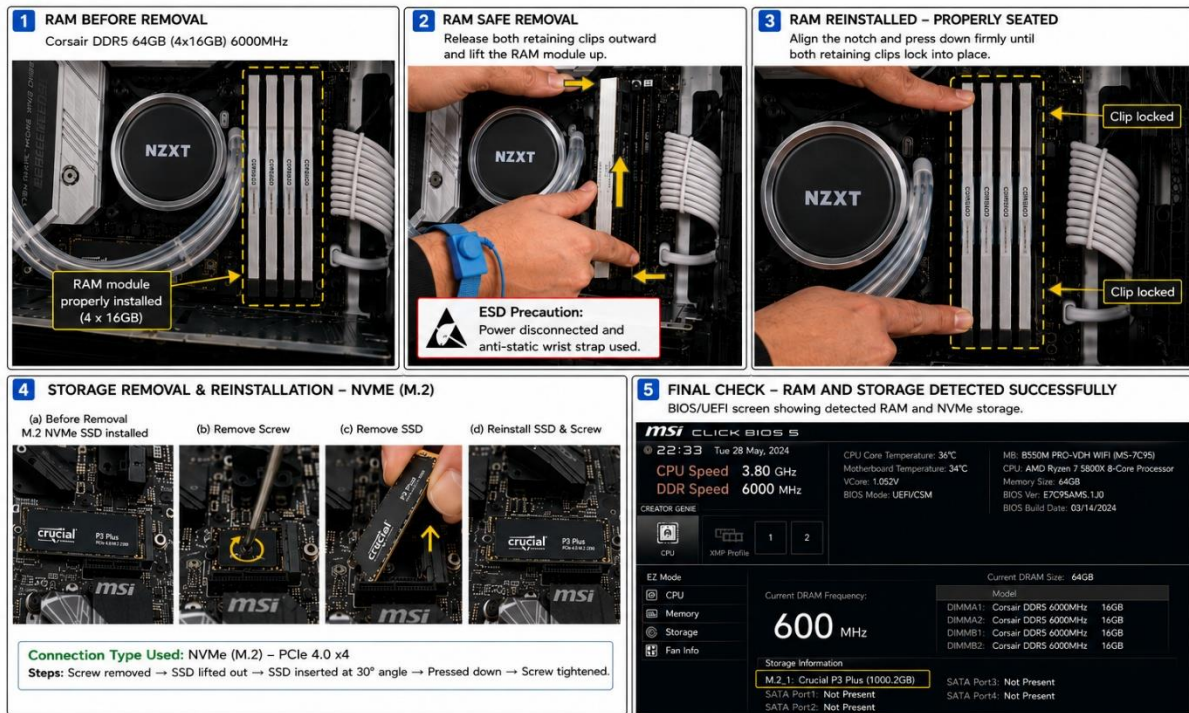


Figure 3: RAM and Storage Removal, Reinstallation and Detection

Note: All components handled with care. ESD precautions followed throughout the process. System powered on after reinstallation and verified in BIOS.

Figure 3: RAM and Storage Removal, Reinstallation and Detection

🔗 Safely remove the RAM module, noting ESD precautions taken.

- As shown in Figure 3 (Panel 2), the system was powered off before removing the RAM. ESD precautions were followed by using an anti-static wrist strap. The retaining clips were released and the RAM module was carefully lifted out of the slot.

3 Reinstall the RAM module correctly and confirm it is seated properly.

- As shown in Figure 3 (Panel 3), the RAM was aligned correctly with the slot and pressed down firmly until the retaining clips locked into place. This confirmed that the RAM was properly seated.

3 Remove and reinstall the storage device, noting the connection type used (SATA/NVMe).

- As shown in Figure 3 (Panel 4), the storage device used was an NVMe M.2 SSD. The retaining screw was removed, the SSD was lifted out, and then it was reinserted at the correct angle and secured with the screw.

3 Remove and reinstall the storage device, noting the connection type used (SATA/NVMe).

- As shown in Figure 3 (Panel 5), the BIOS/UEFI detected the installed RAM and NVMe storage correctly. The system showed 64 GB RAM and the Crucial P3 Plus NVMe SSD, confirming that both components were successfully reinstalled and detected.

Task 4.

❖ Full Desktop PC Assembly.

TASK 4: FULL DESKTOP PC ASSEMBLY



Figure 4: Full Desktop PC Assembly and POST Verification

Note: All components handled with care. Correct sequence followed. ESD precautions maintained throughout the build.

Figure 4: Full Desktop PC Assembly and POST Verification.

🌀 Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU.

- As shown in **Figure 4**, the PC was assembled by installing the **CPU and cooler, RAM, NVMe storage, motherboard, GPU, and PSU**. Each component was fitted into its correct position before connecting the required power cables.

🌀 Follow correct installation sequence and ESD/safety precautions throughout.

- The assembly was completed in a proper sequence, starting with the CPU and cooler, followed by RAM and storage, motherboard installation, GPU

installation, and finally PSU connections. The system was kept powered off during assembly, and ESD precautions were followed while handling the components, as shown in Figure 4.

🔗 Document each step with labeled photos or diagrams.

- **Figure 4** documents the complete assembly process through six labelled steps. It shows the installation of the **CPU and cooler, RAM and storage, motherboard, GPU, PSU connections**, and the final system check.

🔗 Power on the system and confirm it passes POST successfully.

- As shown in **Figure 4, Step 6**, the completed system was powered on and the **BIOS/UEFI screen** was displayed successfully. The system detected the installed components, confirming that **POST was completed successfully**.

✂ Section C – Mini Capstone Project

Task 1.

❖ Mini Project: Workstation Deployment & Boot-Failure Diagnosis.

✚ **Objective:** ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors, and the IT team suspects incorrect installation or compatibility issues. Build (or inspect) a workstation, diagnose the likely causes of these failures, and produce a diagnostic and installation-verification report — combining component selection, installation procedure, and compatibility troubleshooting from Section B.

✚ Project Requirements and Answers:

🌀 **Requirement 1: Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.**

➤ I would first check the **PSU, RAM, and motherboard**.

➤ **PSU:** Check whether the system is receiving power and whether the main power connectors are connected correctly.

➤ **RAM:** Incorrectly seated RAM can prevent the system from completing POST.

- ✦ **Motherboard:** Check for loose connections, incorrect installation, or diagnostic indicators.

These components are checked first because they are important for powering the system and completing the initial boot process.

🔗 Requirement 2: Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.

- The CPU must be compatible with the motherboard's **socket type and supported CPU family**. In the system inspected, the **MSI B550M PRO-VDH WIFI** motherboard uses an **AM4 socket**, which is compatible with the **AMD Ryzen 5 5600X** used in the build.

If the CPU is not supported by the motherboard, the system may fail to boot or complete POST.

🔗 Requirement 3: Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.

- I would check that the **RAM is fully seated** and that its retaining clips are locked. I would also confirm that the **NVMe storage device is correctly inserted into the M.2 slot and secured**.

After installation, I would enter the **BIOS/UEFI** and verify that the RAM and storage device are detected correctly. A successful POST should complete without hardware errors.

🌀 Requirement 4: Produce a hardware-installation checklist of at least three steps to prevent similar failures in future deployments.

- Check **component compatibility** before assembly.
 - Follow **ESD precautions** and ensure all components are properly seated.
 - Check all **power and data connections** before the first boot.
 - Verify hardware detection in **BIOS/UEFI** after assembly.

🌀 Requirement 5: Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.

- For the ABC Solutions lab, I would use **SSDs instead of HDDs** because they provide faster boot times, quicker application loading, and better overall system responsiveness. SSDs also have no moving mechanical parts, making them quieter and suitable for regular office use.

For modern office workstations used regularly for applications and multitasking, an SSD provides a **faster and smoother user experience**.

✂ Section D – Ai Augmented Learning

Task 1.

❖ Step 1 - Build with AI.

➤ Exact Prompt Given to the AI:

Create a simple step-by-step troubleshooting checklist for a junior technician diagnosing a desktop PC that fails to boot immediately after assembly. Arrange the checks in the correct priority order. Cover the motherboard, RAM, storage, and PSU as possible failure points. Keep the checklist easy to follow on-site and apply it to the ABC Solutions scenario, where some newly assembled PCs fail to start or show hardware detection errors.

🧩 AI's Original Checklist:

- Check whether the PC powers on and look for error messages.
- Check the **motherboard** for visible damage or loose connections.
- Check whether the **RAM** is installed correctly and reseal it if necessary.
- Check whether the **storage device** is connected and detected.
- Check the **PSU** cables and power connections.
- Enter **BIOS/UEFI** and check whether the components are detected.
- Try starting the operating system again.

Application to the ABC Solutions Scenario:

In the ABC Solutions lab, the checklist would help identify an incorrectly seated RAM module causing a hardware detection or boot problem. Step 3 would catch this issue by checking and reseating the RAM before continuing with further diagnosis.

❖ Step 1 - Test & Debug (Without AI).

➤ Issue Found in the AI Checklist:

The original checklist does not mention **ESD precautions before handling internal components**. It also starts checking components without clearly stating that the system should be **powered off and disconnected from power** first.

Corrected Checklist:

- **Power off and unplug the PC** and follow proper **ESD precautions** before opening the case.
- Check the **PSU** and confirm that the 24-pin motherboard, CPU, and GPU power connectors are properly connected.
- Check the **RAM** and make sure it is fully seated with the retaining clips locked.
- Check the **motherboard** for loose connections, incorrect installation, or diagnostic indicators.
- Check the **storage device** and confirm that the SATA or M.2 connection is secure.
- Power on the system and check **POST/BIOS/UEFI** for hardware detection or error messages.

➤ If the hardware is detected correctly, continue with the operating system boot check.

✚ What I Changed and Why:

I added **ESD precautions and power disconnection as the first step** because internal components should not be handled while the system is powered. I also placed the physical hardware checks before software-level checks. This makes the troubleshooting process safer and more logical for a junior technician.